

CSE321: Computer Networks

Class Test – 1

Student #: _____

Time: 10 minutes

Mark: 20

[Instructions: 1) This is an **OPEN-MATERIAL** examination. 2) Type your answer in the place after "**Answer:**" below. 3) Then, make a pdf file of it (your answer) and submit that (pdf) in Moodle *in time*. 4) Check and confirm at your own that your submitted file is okay and not corrupted. Grading will be made based on what will be available in Moodle.]

Question:

You and your group-mate are given a task of designing a DLL protocol for data communication of a very sensitive military application. In this regard, you and your group-mate are explicitly asked to perform the task of error detection in your developed DLL protocol.

To do so, your group-mate proposes to adopt the following aspects in your developed DLL protocol –

1. Byte stuffing for framing,
2. r parity bits for m data bits maintaining the condition that $(m+r+1) \leq 2^r$ along with arranging k consecutive codewords in a matrix and transmitting one column at a time to handle a maximum burst error of k bits, and
3. Sliding window protocol using Selective Repeat with a window size of $(MAX_SEQ + 1)/2$.

Now, you need to pinpoint what your group-mate has done wrong in the above proposal from the perspective of performing the task of error detection. Justify your answer.

Answer to the point, as wrong answer or wandering here and there could result in negative point(s).

Answer: